

DEPARTMENT OF ROBOTICS & ARTIFICIAL INTELLIGENCE

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Programming for Artificial Intelligence LAB

Project Report

Topic: Mobile Price Prediction

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Submitted To: **Mam Zahra Khalid**

Submitted By:

Student Names	Registration Numbers
Syeda Nalain Zainab Bukhari	24108233
Tehniat Khan	24108235
Aiman Nasir	24108202
Malika Naz	24108212
Safia Alam	24108229

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Topic: Mobile Price Prediction using Pandas

1. Overview:

The Student Performance Analysis System is a Python-based project developed to analyze and preprocess student data using Pandas and Matplotlib libraries.

In this project, a dataset is used and cleaned through different data preprocessing techniques. The dataset contains student information such as names, study hours, attendance and marks.

The system performs operations such as:

- Checking missing values
- Filling incomplete data
- Cleaning the dataset
- Statistical analysis
- Data visualization

Pandas library is used for dataset preprocessing and cleaning while Matplotlib is used to create graphs and visual representations of the data.

Data preprocessing is one of the most important steps in Artificial Intelligence and Machine Learning because raw datasets often contain incomplete or unorganized data. Therefore, this project demonstrates how preprocessing techniques can improve data quality before analysis.

Overall, this project shows how Python libraries such as Pandas and Matplotlib can be used together for efficient data analysis and preprocessing.---

2. Why this topic:

The Student Performance Analysis System was selected as the project topic to demonstrate how datasets can be cleaned and preprocessed using Pandas library.

The main reason for choosing this topic is to understand how real-world datasets are managed before applying Artificial Intelligence and Machine Learning techniques.

Most datasets contain:

- Missing values
- Duplicate data
- Incorrect information
- Unorganized records

Therefore, preprocessing is required to clean and organize the data properly.

This topic also allows the implementation of important concepts such as:

- Dataset handling using Pandas
- Data preprocessing
- Missing value handling
- Statistical analysis
- Data visualization using Matplotlib

Furthermore, the project helps students understand how clean and organized datasets improve the efficiency and accuracy of AI systems.

Overall, this topic was selected because it demonstrates practical implementation of Pandas-based preprocessing techniques used in Artificial Intelligence.

3. Features:

*** Dataset Input System:**

Stores student records such as names, marks, attendance and study hours.

*** Missing Value Detection:**

Identifies missing values in the dataset.

*** Data Cleaning:**

Handles missing values using mean replacement.

*** Statistical Summary:**

Displays dataset statistics using Pandas.

*** Data Visualization:**

Generates graphs using Matplotlib.

*** Bar Chart Generation:**

Displays student marks comparison.

*** Histogram Visualization:**

Shows marks distribution.

*** Pie Chart Representation:**

Displays percentage contribution of student marks.

*** Structured Data Processing:**

Uses Pandas DataFrames for organized data handling.

*** Easy Data Analysis:**

Allows efficient understanding of student performance trends.

4. Modules:

- Input Module:

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- Handles student dataset creation and input storage.
- Data Processing Module:
 - Processes and cleans dataset using Pandas functions.
- Missing Value Module:
 - Detects and fills missing values in the dataset.
- Statistical Analysis Module:
 - Generates statistical summaries of the dataset.
- Visualization Module:
 - Creates graphical charts using Matplotlib.
- Output Module:
 - Displays cleaned dataset and graphs to the user.

5. Purpose:

The purpose of the Student Performance Analysis System is to demonstrate how data preprocessing and analysis are performed using Python libraries.

The system processes student records, cleans incomplete data and generates visual reports for analysis.

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The project also aims to show how Pandas and Matplotlib can be used together for efficient dataset management and visualization.

6. Objectives:

- * To develop a Student Performance Analysis System using Python.
- * To apply Pandas for dataset preprocessing and cleaning.
- * To detect and handle missing values in datasets.
- * To perform statistical analysis on student records.
- * To generate visual charts using Matplotlib.
- * To understand the importance of clean data in AI systems.
- * To build a structured and user-friendly preprocessing system.

7. Methodology:

The Student Performance Analysis System follows a structured preprocessing approach for analyzing student data.

First, the dataset is created using Pandas DataFrames. The dataset contains student names, study hours, attendance and marks.

The system then checks dataset dimensions and identifies missing values using Pandas functions.

Missing values are handled by replacing them with average values using mean calculations.

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After preprocessing, statistical summaries are generated using describe() function to analyze dataset information.

Finally, Matplotlib is used to generate graphs such as bar charts, histograms and pie charts for better visualization of student performance.

8. Scope:

The scope of the Student Performance Analysis System is limited to preprocessing and analysis of student datasets using Python libraries.

The system is designed mainly for educational purposes to demonstrate data preprocessing concepts used in Artificial Intelligence and Machine Learning.

The project can be extended in the future by adding:

- **Real CSV datasets**
- **Machine Learning models**
- **GUI interfaces**
- **Advanced data analytics**
- **Database integration**

9. Benefits of the system:

- * **Simplifies dataset preprocessing**
- * **Helps identify missing values easily**
- * **Improves dataset quality**
- * **Demonstrates practical implementation of Pandas**

- * Provides graphical representation of data

- * Makes data analysis easier and more organized

- * Can be extended for Machine Learning applications

10. Limitations:

- * Uses a small sample dataset

- * No database integration

- * Limited visualization features

- * No graphical user interface

- * Only basic preprocessing techniques are implemented

- * Does not perform Machine Learning predictions

11. Tools and Techniques used:

Programming Language:

Python is used to develop the complete system.

Pandas Library:

Used for dataset preprocessing, cleaning and analysis.

NumPy Library:

Used for numerical calculations and missing value handling.

Matplotlib Library:

Used for data visualization and graphical representation.

Data Preprocessing:

Used to clean and organize raw data.

Statistical Analysis:

Used to analyze student performance information.

12. Conclusion:

The Student Performance Analysis System demonstrates how data preprocessing and analysis techniques can be applied using Python libraries such as Pandas and Matplotlib.

The system successfully implements dataset cleaning, missing value handling, statistical analysis and graphical visualization.

Although the project is simple and developed for educational purposes, it provides a clear understanding of how

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preprocessing is performed before applying Artificial Intelligence and Machine Learning techniques.

Future improvements may include larger datasets, machine learning integration and advanced data visualization techniques.

13. Work Distribution:

Name	Task
Syeda Nalain Zainab Bukhari	Dataset collection and project documentation
Tehniat Khan	Data preprocessing and missing value handling using Pandas
Aiman Nasir	Dataset analysis and statistical summary generation
Malika Naz	Data visualization using Matplotlib graphs and charts

14. CODE:

```
# =====  
  
# MOBILE PRICE PREDICTION MODEL  
  
# Using Pandas + Scikit-Learn + Matplotlib  
  
# Dataset Loaded Directly From Website  
  
# =====  
  
# Install Required Libraries:  
  
# pip install pandas scikit-learn matplotlib seaborn  
  
# =====  
  
# IMPORT LIBRARIES
```

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```
# =====
```

```
import pandas as pd
```

```
import numpy as np
```

```
import matplotlib.pyplot as plt
```

```
import seaborn as sns
```

```
from sklearn.model_selection import train_test_split
```

```
from sklearn.preprocessing import StandardScaler
```

```
from sklearn.linear_model import LinearRegression
```

```
from sklearn.metrics import mean_absolute_error
```

```
from sklearn.metrics import mean_squared_error
```

```
from sklearn.metrics import r2_score
```

```
# =====
```

```
# LOAD DATASET FROM WEBSITE
```

```
# =====
```

```
url = "https://raw.githubusercontent.com/amankharwal/Website-data/master/mobile_prices.csv"
```

```
data = pd.read_csv(url)
```

```
# =====
```

```
# DISPLAY DATASET
```

```
# =====
```

```
print("\nFIRST 5 ROWS")
```

```
print(data.head())
```

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```
print("\nDATASET SHAPE")
```

```
print(data.shape)
```

```
print("\nMISSING VALUES")
```

```
print(data.isnull().sum())
```

```
# =====
```

```
# VISUALIZATION
```

```
# =====
```

```
plt.figure(figsize=(12, 8))
```

```
sns.heatmap(data.corr(),
```

```
            annot=True,
```

```
            cmap="coolwarm")
```

```
plt.title("Correlation Heatmap")
```

```
plt.show()
```

```
# =====
```

```
# FEATURES AND TARGET
```

```
# =====
```

```
# Target = price_range
```

```
X = data.drop("price_range", axis=1)
```

```
y = data["price_range"]
```

```
# =====
```

```
# FEATURE SCALING
```

```
# =====
```

```
scaler = StandardScaler()
```

```
X_scaled = scaler.fit_transform(X)
```

```
# =====
```

```
# SPLIT DATA
```

```
# =====
```

```
X_train, X_test, y_train, y_test = train_test_split(
```

```
    X_scaled,
```

```
    y,
```

```
    test_size=0.2,
```

```
    random_state=42
```

```
)
```

```
# =====
```

```
# TRAIN MODEL
```

```
# =====
```

```
model = LinearRegression()
```

```
model.fit(X_train, y_train)
```

```
# =====  
  
# MAKE PREDICTIONS  
  
# =====  
  
y_pred = model.predict(X_test)  
  
# =====  
  
# EVALUATE MODEL  
  
# =====  
  
mae = mean_absolute_error(y_test, y_pred)  
mse = mean_squared_error(y_test, y_pred)  
r2 = r2_score(y_test, y_pred)  
  
print("\nMODEL PERFORMANCE")  
print("-----")  
  
print("Mean Absolute Error :", mae)  
print("Mean Squared Error :", mse)  
print("R2 Score      :", r2)  
  
# =====  
  
# TAKE USER INPUT  
  
# =====  
  
print("\n==== MOBILE PRICE PREDICTION ====")
```

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```
battery_power = int(input("Battery Power: "))
```

```
ram = int(input("RAM (MB): "))
```

```
px_height = int(input("Pixel Height: "))
```

```
px_width = int(input("Pixel Width: "))
```

```
mobile_wt = int(input("Mobile Weight: "))
```

```
int_memory = int(input("Internal Memory (GB): "))
```

```
talk_time = int(input("Talk Time: "))
```

```
# =====
```

```
# CREATE USER DATA
```

```
# =====
```

```
user_data = [[
```

```
    battery_power,
```

```
    0,          # blue
```

```
    1,          # clock_speed
```

```
    1,          # dual_sim
```

```
    1,          # fc
```

```
    0,          # four_g
```

```
    int_memory,
```

```
    1,          # m_dep
```

```
    mobile_wt,
```

```
    1,          # n_cores
```

```
    1,          # pc
```

```
    px_height,
```

```
    px_width,
```

```
    ram,
```

```
    1,          # sc_h
```

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```
1,      # sc_w

talk_time,

1,      # three_g

1,      # touch_screen

1      # wifi

]]

# =====

# SCALE USER DATA

# =====

user_scaled = scaler.transform(user_data)

# =====

# PREDICT PRICE RANGE

# =====

prediction = model.predict(user_scaled)

predicted_class = round(prediction[0])

# =====

# DISPLAY RESULT

# =====

print("\n=====")

if predicted_class <= 0:
```

```
print("Predicted Price Range: LOW COST MOBILE")
```

```
elif predicted_class == 1:
```

```
    print("Predicted Price Range: MEDIUM COST MOBILE")
```

```
elif predicted_class == 2:
```

```
    print("Predicted Price Range: HIGH COST MOBILE")
```

```
else:
```

```
    print("Predicted Price Range: VERY HIGH COST MOBILE")
```

```
print("=====")
```

```
# =====
```

```
# VISUALIZE ACTUAL VS PREDICTED
```

```
# =====
```

```
plt.figure(figsize=(8, 5))
```

```
plt.scatter(y_test,  
            y_pred,  
            color="green")
```

```
plt.xlabel("Actual")
```

```
plt.ylabel("Predicted")
```

```
plt.title("Actual vs Predicted Price Range")
```

